

DNS Resolver Recommender

Measuring & Comparing DNS over HTTPS Performance in the Browser

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Course: Trends in Internet Measurements

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Create Knowledge.**

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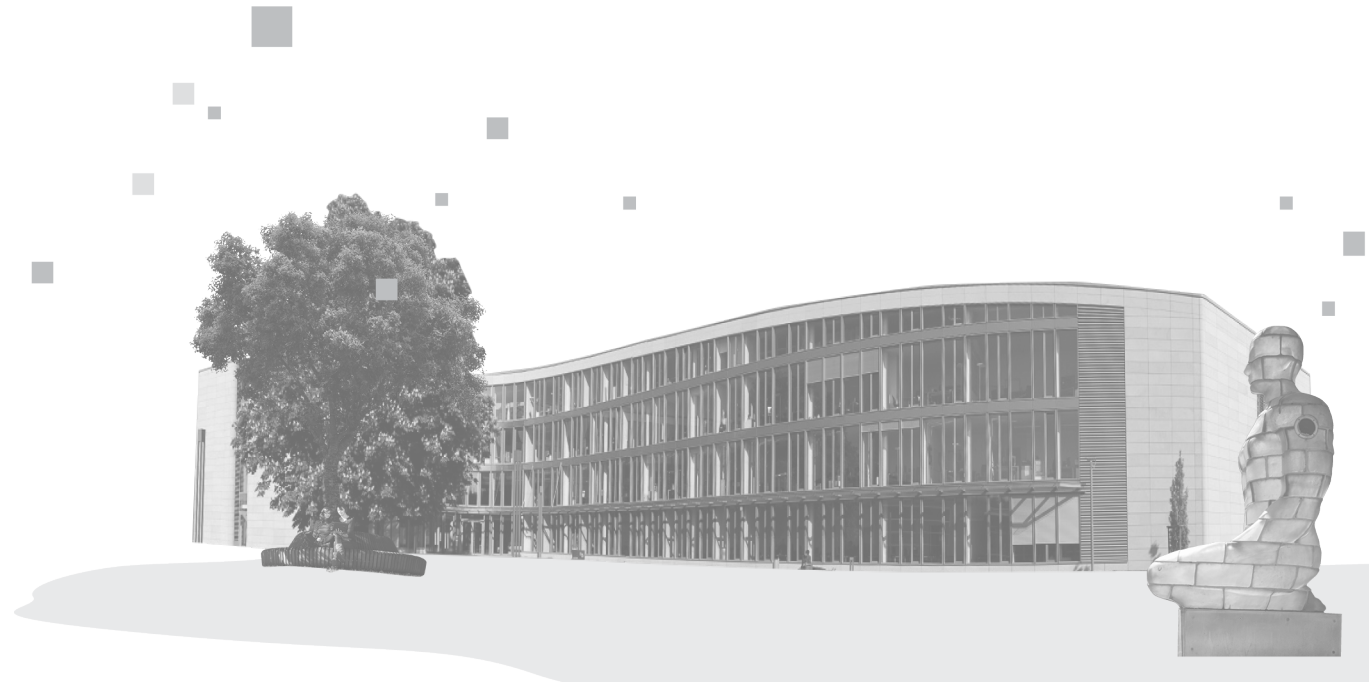


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Motivation: Why DNS Resolvers Matter



Speed/Latency

Modern web pages require dozens of DNS lookups



Privacy

ISPs monetize browsing data [7]



Security

Public resolvers offer malware and ad-blocking

"Many ISPs collect and share far more data about their customers than consumers expect..." (FTC, 2021).

[7] FTC Report on ISP Data Monetization

Problem Statement: The Blind Spot



Network Engineering Reality

- Resolvers like 1.1.1.1 (Cloudflare) or 9.9.9.9 (Quad9) exist.
- Optimal choice depends heavily on:
 - Geography
 - ISP Routing
 - Time of day

The Average User Reality

- Blindly trusts the ISP default.
- No easy way to measure performance locally.
- Existing CLI tools are too complex for non-technical users.

State of the Art: Desktop & CLI Tools



Desktop & Network Engineering Tools

- **GRC Benchmark / namebench (Google)**
 - *Protocol:* UDP (Port 53)
 - *Depth:* High
 - *Barrier:* High (Requires Installation)
- **CLI Tools (e.g., dnsperf)**
 - *Protocol:* UDP/TCP/DoH
 - *Depth:* Very High
 - *Barrier:* Very High
- **Data Collection:** None (No global telemetry).

```
.101, 142.251.127.139
8.8.8.8      8.8.8.8      592 ms | www.google.com is hijacked: 142.251.157.119, 14
2.251.156.119, 142.251.153.119, 142.251.154.119, 142.251.152.119, 142.251.151.119, 142.251.
155.119, 142.251.150.119, google.com appears incorrect: 142.251.39.238, twitter.com appears
incorrect: 162.159.140.229, www.google-analytics.com appears incorrect: 142.251.142.46, st
atic.ak.fbcdn.net.: Timeout

- Sending 10 queries to 3 servers: 0/30.....14.....21.....29.30/30
- Error querying 8.8.8.8 [8.8.8.8]: auction1.taobao.com.: Timeout

Fastest individual response (in milliseconds):
-----
1.1.1.1      ##### 25.30003
8.8.8.8      ##### 58.87079

Mean response (in milliseconds):
-----
1.1.1.1      ##### 71.81
9.9.9.9      ##### 219.69
8.8.8.8      ##### 433.33

Response Distribution Chart URL (200ms):
-----
http://chart.apis.google.com/chart?cht=lx&chs=720x415&cht=x,y&chg=10,20&chxr=0,200|1,0
,100&chdt=0,13,15,16,32,37,37,51,52,52|0,10,20,30,40,50,60,80,90,100|0,29,37,42,45,51,60,17
50|0,10,20,30,40,80,90,100|0,6,6,17,20,51,72,819|0,10,30,40,50,80,90,100&chco=ff9900,1a00ff
,ff00e6&chxt=x,y,x,y&chxl=2:|Duration+in+ms|3:|[%25]&chdl=1.1.1.1|8.8.8.8|9.9.9.9

Response Distribution Chart URL (Full):
-----
http://chart.apis.google.com/chart?cht=lx&chs=720x415&cht=x,y&chg=10,20&chxr=0,9,3500|1,0
,100&chdt=0,1,1,1,2,2,2,3,3,3|0,10,20,30,40,50,60,80,90,100|0,2,2,2,3,3,3,100|0,10,20,30,4
0,80,90,100|0,0,0,1,1,3,4,47|0,10,30,40,50,80,90,100&chco=ff9900,1a00ff,ff00e6&chxt=x,y,x,y
&chxl=2:|Duration+in+ms|3:|[%25]&chdl=1.1.1.1|8.8.8.8|9.9.9.9

Recommended configuration (fastest + nearest):
-----
nameserver 1.1.1.1      # 1.1.1.1
nameserver 9.9.9.9      # 9.9.9.9
nameserver 8.8.8.8      # 8.8.8.8

*****
In this test, 1.1.1.1 is N/A:
*****

- Saving report to /tmp/namebench 2026-04-29 1756.html
- Saving detailed results to /tmp/namebench 2026-04-29 1756.csv
[noaa@archlinux ~]$
```

Typical CLI-based DNS measurement tool

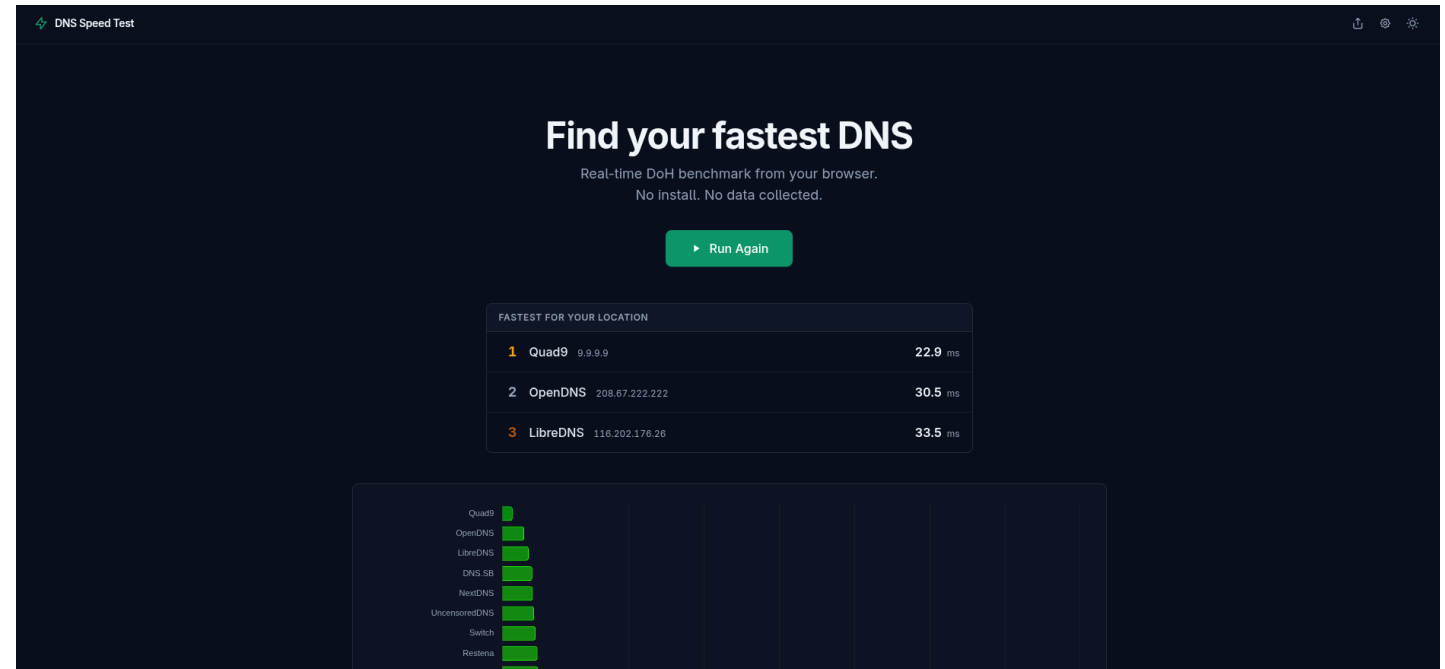
Google (2010). namebench.

Browser Tools

- **Example:** DoHSpeedTest
- *Protocol:* DoH via JS fetch()
- *Barrier:* Low
- *Depth:* **Low (JS timing only)**

Try it yourself:

<https://dnsspeedtest.online/>



The DoHSpeedTest web interface

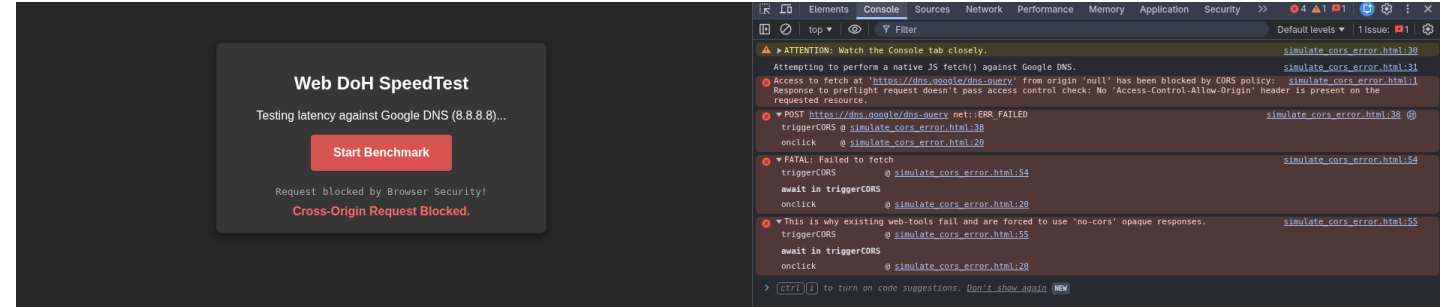
[3] BrainicHQ: DoHSpeedTest (GitHub)

State of the Art: The CORS Problem



The Flawed Methodology

- Fails on major resolvers (e.g., Google, Quad9) due to strict CORS policies.
- Forced no-cors mode creates "opaque responses".
- JS cannot parse binary DNS responses.
- Treats the resolver as a black box (Cannot verify valid IP vs NXDOMAIN).



Typical CORS violation when querying public DNS via JavaScript

CORS Security Mechanism in Modern Browsers

Proposed Solution: The DNS Resolver Recommender



Key Innovations

- **WebAssembly Core:** Bypasses JS limitations.
- **DoH:** Tunnels DNS through browser sandbox.
- **Dual-Metric:** Cached vs uncached latency.
- **Privacy:** Ephemeral processing, no IP logging.



Proposed user interface (Simple vs Expert Mode)

Research Questions & Design Goals



Research Questions:

1. How can we do DNS in the web? (JS, JS Libs, WASM, ...?)
2. What protocols can we use for DNS in the web?
3. How can we do cached vs. uncached DNS in the web?
 - *Approach: Custom authoritative DNS resolver*
4. How to collect user data?

Design Goals:

1. Setup a website that compares popular DNS recursive resolvers
2. Each measurement collects data

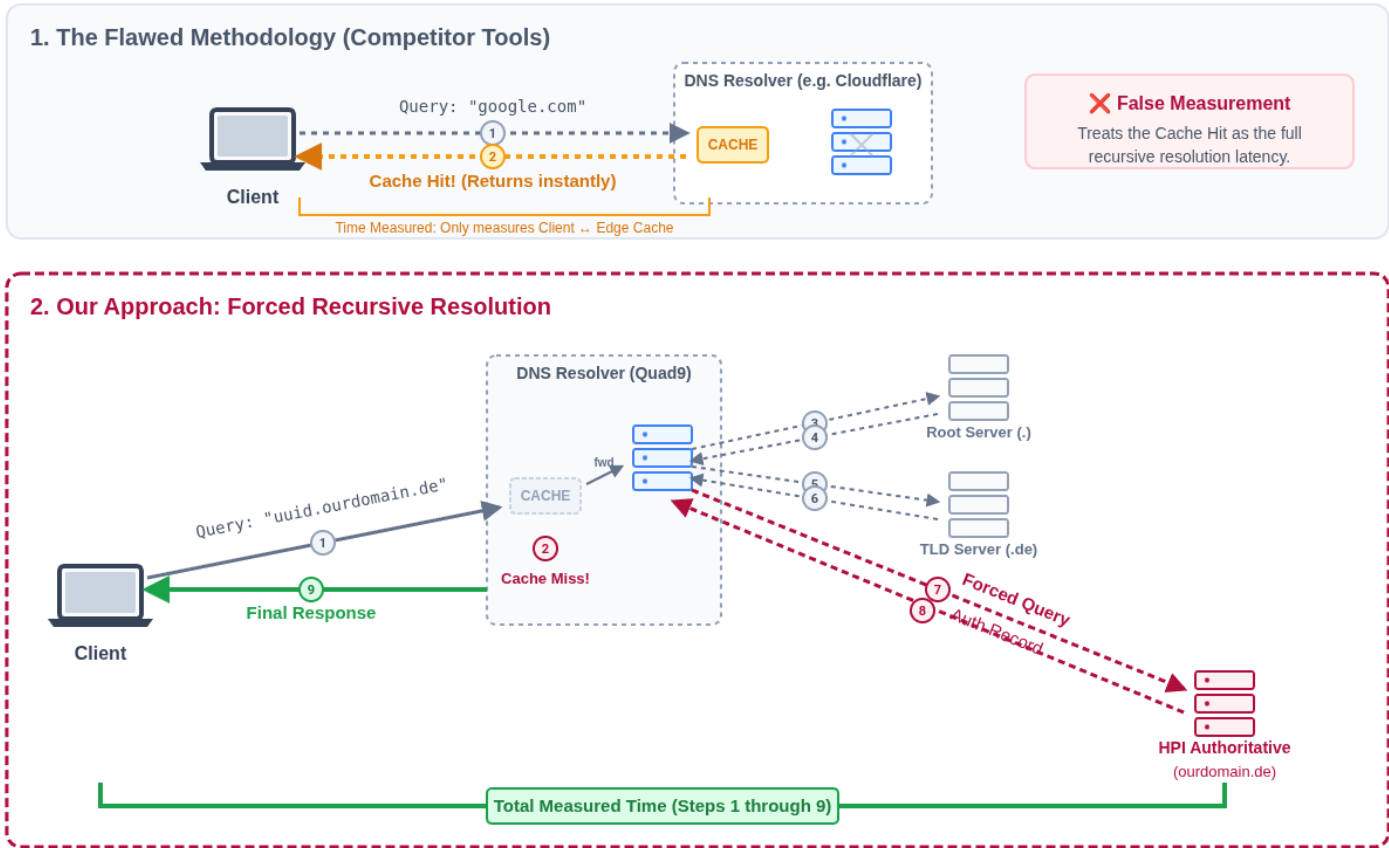
Bypassing the Browser Sandbox

- **RQ2 (Protocols):** Native UDP/TCP Port 53 is **Blocked**. Solution: DNS-over-HTTPS (DoH, RFC 8484) via Fetch API (Port 443).
- **RQ1 (Execution):** How to process raw application/dns-message bytes efficiently?
 - Pure JS: Inefficient bit-level manipulation.
 - **Wasm Bridge:** Rust hickory-dns compiled to WebAssembly securely parses binary packets at near-native speeds.

RQ3: Cached vs. Uncached DNS in the Web

Overcoming the Cache

- Browsers and OS caches distort measurements.
- We deploy a **Custom Authoritative DNS Server**.
- Generates unique UUID subdomains.
- Forces the public resolver to perform a full recursive lookup.



Isolating True Recursive Processing vs Edge Caches

[2] Borgolte et al.: How DNS over HTTPS is reshaping privacy

RQ4: Collecting User Data (Privacy)



Data Collection & Privacy Compliance

- Dynamic IP addresses are Personal Data (CJEU Breyer Case, 2016).
- Strict adherence to GDPR / DSGVO (Data Minimization - Art. 5c).
- **Primary Solution:** Ephemeral RAM Processing. Map IP → ASN/GeoIP → Instantly drop IP.
- **Alternative / Future Solution:** Oblivious DNS over HTTPS (ODoH).

[4] CJEU: Patrick Breyer v Bundesrepublik Deutschland

System Architecture



1. Client-Side (Browser)

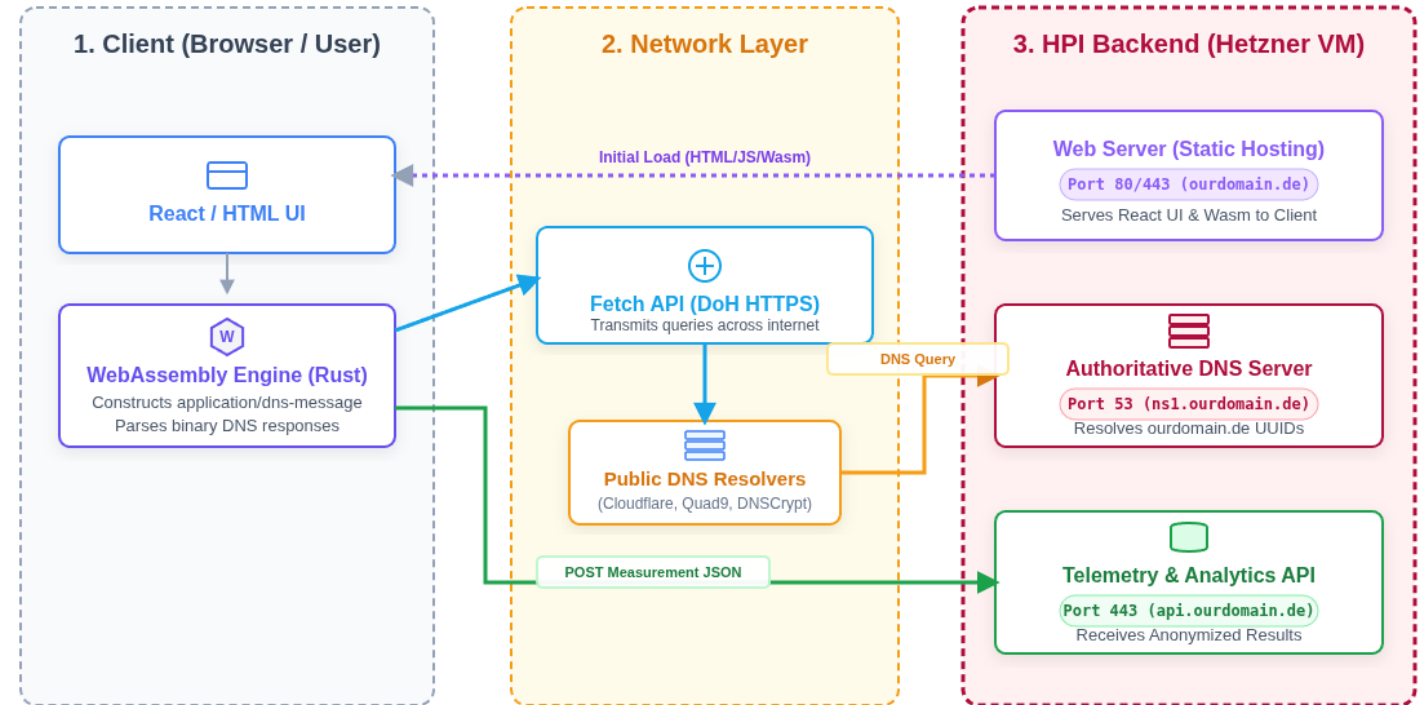
- React/Vue UI
- WebAssembly Engine (Rust)

2. Network Layer

- DoH Fetch API
- Public Resolvers

3. Backend (Hetzner VM)

- Port 53: Authoritative DNS
- Port 443: Telemetry API



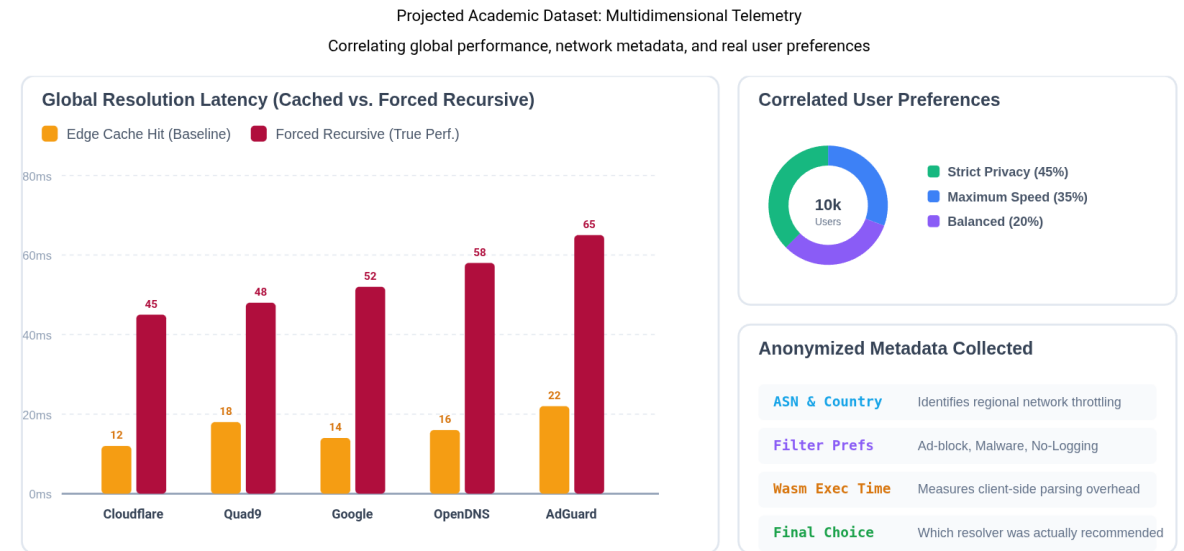
High-level architectural flowchart

Expected Outcomes & Timeline



Timeline

- **Step 1 (Foundation):** Deploy Server on Hetzner VM & finalize Wasm Client.
- **Step 2 (Core Logic):** Implement Cached/Uncached logic and Recommendation Engine.
- **Step 3 (Refinement):** Telemetry API, Privacy compliance, and UI polishing.



Projected dashboard mapping latencies, metadata, and user preferences

Future Work: Beyond the Initial Web App



Evolution Paths

- **Form Factor:** Evolution from a one-time Web App to a continuous Browser Extension for longitudinal data.
- **Protocol Expansion:** DNS over QUIC (DoQ) integration for lower latencies.
- **Advanced Recommender:** Machine Learning for real-time routing predictions instead of static thresholds.

Thank You!
Questions?

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